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DATA MINING, MODELING AND ANALYTICS ASSIGNMENT

PROJECT 1: BANK MARKETING CLASSIFICATION

1. Introduction

This project investigated the use of machine learning to predict whether a bank customer would subscribe to a term deposit following a marketing campaign. The analysis used the Bank Marketing dataset, which contains customer demographic, financial and campaign-related information. The objective was to develop and evaluate classification models that could help identify potential subscribers.

2. Data Preparation and Exploration

The original dataset contained 41,188 records and 21 columns. Initial exploration identified 12 duplicate records, which were removed, leaving 41,176 observations. No missing values were found, although some categorical fields contained the value **"unknown".** The target variable was imbalanced: approximately 88.73% of customers did not subscribe, while 11.27% subscribed.

The `**duration**` variable was excluded from modelling because call duration is only known after a marketing call and would not be available when selecting customers beforehand. The remaining data was divided into training and testing sets using an 80:20 stratified split.

3. Modelling and Evaluation

Three classification algorithms were evaluated: Logistic Regression, Random Forest and HistGradientBoosting. A majority-class Dummy Classifier was also used as a baseline. Precision, recall, F1-score and ROC-AUC were used to assess model performance, with particular attention to recall because identifying potential subscribers was an important objective.

Logistic Regression achieved a recall of 64.55% and an F1-score of 46.15%. Random Forest achieved an F1-score of 45.43%, while HistGradientBoosting recorded the highest accuracy (90.10%), precision (66.67%) and ROC-AUC (0.8131). The baseline achieved an accuracy of 88.73% but failed to identify any actual subscribers, demonstrating why accuracy alone was insufficient for this imbalanced dataset.

Threshold analysis was conducted using validation data. A threshold of 0.60 produced the highest validation F1-score of 48.99%. On the test set, the selected Logistic Regression model achieved 41.77% precision, 62.39% recall and an F1-score of 50.04%.

4. Conclusion

Logistic Regression with a decision threshold of 0.60 was selected as the recommended model because it provided a useful balance between identifying potential subscribers and limiting unnecessary marketing calls. However, the model still produced false positives and missed some actual subscribers.

The findings demonstrate the potential of machine learning to support bank marketing decisions. Nevertheless, the results should be interpreted with consideration of class imbalance and possible changes in customer behaviour over time. The model identifies patterns in historical data and does not guarantee future customer responses.